

Knowledge Model for Developing, Searching and Using Personalized Learning Content for Learners, Having Dyslexia Disability

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ABSTRACT

Much research in the area of e-learning systems today is focused on personalization and adaptivity, but only a few of the proposed systems are intended for students, having learning disabilities. These learners are significant amount of all learners (children, having some dyslexia symptoms are between 10% and 20% of all children). Dyslexia symptoms can be specific for different learners and can require specific organization of learning. All the experts in learning say that suitable learning is a key for achieving excellent results for these learners.

We made extensive study on the relevant theories intended for better understanding of the requirements of an e-learning for people, having learning disabilities and existing frameworks or tools for people with dyslexia. As a result we propose a knowledge model for developing, describing, searching and recommending personalized learning resources in the web for learners, having dyslexia learning disability. We also propose agent-based software architecture, that manage and use implemented knowledge for supporting leaning resource development, searching, ranking according to specified criteria and recommendation of these resources to dyslexic learners.

CCS CONCEPTS

•General and reference → Document types → Surveys and overviews; • Information systems → Data management systems; •Computing methodologies → Artificial intelligence → Knowledge representation and reasoning

KEYWORDS

Ontology, Dyslexia, E-Learning, Web-based learning, knowledge

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model, personalized learning.

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1 Introduction

There are grand amount of resources, useful for learning. Finding the best ones for specific learner that correspond to its goals is of great importance for successful learning. Usually resources are searched and retrieved by usage of computer programs, as search engines or educational systems. It is difficult for computer programs to understand e-learning content and quality, and its knowledge about potential learners is very restricted. So, computer-readable semantic descriptions of resources learners and learning process are needed. In adaptive e-learning systems the learners are categorized based on various properties (for example, their proficiency, preferences, learning styles or disabilities) [1], [2], [3] for providing personalized learning or assessment content. Many suitable learning resources have been developed having in mind psychological and cognitive abilities of various categories learners, as well as their technical experience and cultural background. One of the most important challenges in e-learning is to realize adaptive personalization according to the learner's changing behavior.

Semantic web technologies as ontological representation of knowledge can make knowledge about learners, learning domains, pedagogy, etc. understandable and usable for computer programs. Personalization of e-learning system can be achieved by using Semantic web technologies. Personalization is good for every learner, but it is very important for peoples, having learning disabilities, as dyslexia, because special types of learning resources are needed for supporting its successful learning.

Dyslexia is a language-based learning disability. People with dyslexia have difficulties in spelling, reading, and writing, and in

this way learning is more difficult for them. Dyslexics can achieve high results in learning, science, and in every other area, but they need from specific type adaptive learning to overcome some learning problems. High quality multimodal learning resources meeting specific presentation requirements can make learning easier and more effective for dyslexics. On the other hand, dyslexics as other learners have its learning styles, preferences, etc. So, needed adaptation of learning content for dyslexics should be more flexible and highly specialized.

Over the past decade there is significant scientific interest on usage of ontologies in e-learning. To make some brief evaluation of dynamics of this interest, we sent queries to google scholar and specialized digital libraries as Scopus, ACM, IEEE, and count papers, returned as relevant to these queries. Some of our queries where: “+dyslexia”, “+ontology +dyslexia +agents”, “+ontology +e-learning”, “+ontology +e-learning + dyslexia”. Examples of our evaluation results are shown on fig 1 and fig.2. Our conclusions are that while scientific interest to the usage of ontologies in e-learning is significant, but nearly constant during last 10 years, the interest to usage of ontologies in e-learning for dyslexics is growing. So, usage of ontologies for personalization of learning for peoples with dyslexia is a hot research area.

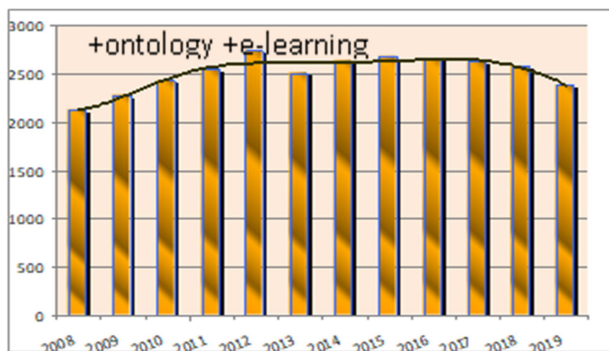


Figure 1: Diagram of the number of search results in Google Scholar, containing both words ‘ontology’ and ‘e-learning’

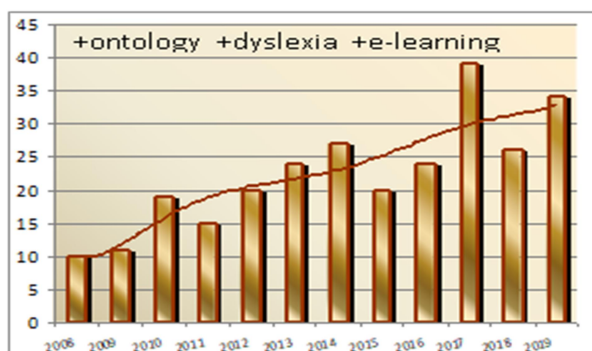


Figure 2: Diagram of the number of search results in google scholar, containing all words ‘ontology’, ‘dyslexia’, ‘e-learning’

Smart learning environments need to consider advanced data manipulation techniques such as employing big data and learning

analytics to collect, combine and analyze individual learning profiles (from past to present) in order to scientifically generalize and infer each individual learning need in real time in ubiquitous settings that encompass both physical and online activities. Learning analytics by using big data can monitor individual learners’ progress and behavior continuously in order to explore factors that may influence learning efficiency and effectiveness [4]. However, providing each learner with just in time adaptive learning support through effective application of full context awareness is still a challenge [5].

2 Related work

Learning management systems (LMS), such as WebCT, EDU2O, Blackboard, or Moodle, are widely used e-learning systems. There are several researches on personalization in these systems, but in practice, individual needs of students do not take into consideration. There are only a few researches on development of adaptive e-learning systems for supporting learning for students with learning difficulties. The use of domain ontology for developing assistive learning technologies for learners with dyslexia is examined in [6]. Two characteristics were employed within this research to describe each resource: dyslexia type and the features of assistive technologies that were deemed to be most appropriate for educational experiences targeted at each dyslexia type. Authors said that the ontology was evaluated in accordance with the needs of students who suffer from dyslexia but results are not presented. [2] Proposes a semantic approach for dynamic generation of personalized learning materials using ontology-based engine that composes and adapts learning materials according to learner’s dyslexia type and learning styles. The approach aims to improve flexibility, extensibility and reusability of resources, and to offer a pedagogically effective and satisfactory learning experience for learners with dyslexia. The flexibility of the approach is not sufficient. It classify resources according to dyslexia type and cannot propose cannot propose specific resources for dyslexia symptom or symptoms combination.

The Semantic web-based educational system that includes software agents, ontology and learning objects is presented in [3]. Software agents are employed to monitor the learner’s actual learning style and modify them accordingly. The learner’s learning style and their modifications are made within the proposed e-learning system. Cloud storage is used as the primary back-end in order to maintain the ontology, databases and other required server resources. The system has modern architecture, but is not intended for learners, having learning disabilities.

Motivation is very important for dyslexics, as they have serious troubles during learning. System architecture for motivation-enhanced learning is proposed in [7]. The construction of the motivation model use ontological approach. The proposed framework of the personalized learning system incorporates a motivation model and corresponding personalization mechanism aiming to improve learning motivation and performance of students with dyslexia.

Usage of ontologies in the process of text simplification to make scientific and learning content more accessible for peoples, having dyslexia, or other learning disabilities is discussed in [8]. The author analyzes many recent researches, related to text simplification and Controlled Natural Languages and its application for development of textual learning resources for dyslexic learners in several languages.

[9] Proposes the use of ontologies to represent accessibility needs and preferences of learners in order to structure the knowledge and to access the information for recommendations and adaptations in e-learning contexts. The dyslexia knowledge is not included in this model, but the ontology can easily be extended.

The possibilities to use virtual learning environments (VLEs) to satisfy the needs of learners with multiple disabilities are discussed in [10]. The proposed VLE employs AI to show how specific learning materials from a huge repository of learning materials can be recommended to learners with various disabilities. Simple logical operators, knowledge based rules, and machine learning based rule induction are combined in this approach.

There are many multimedia teaching and learning course-ware for dyslexic children to learn English but none of the courseware takes into account the cross-cultural specifics and phonological difficulties faced by individuals with dyslexia. The study in [11] examines a framework that could integrate cultural and linguistic factors for Malaysian dyslexic English language learners.

A personalized assessment system in an E-learning is proposed in [1]. The paper presents an ontology-based approach to develop a personalized e-learning which creates adaptive content based on learner's abilities, learning style, level of knowledge and preferences. Ontology is used to represent the content, learner and domain models. The system recognizes changes in the learner's level of knowledge as they progress. Only theoretical approach of designing a e-learning system for dyslexics is discussed in this work.

There are three different approaches for adapting learning content for dyslexics according to [12]

- a) Multidimensional;
- b) Inferential;
- c) Hierarchal.

The characteristics of multidimensional approach include inclination towards complex structure and designing (including audio, video, or other high quality multimedia content). Characteristics of inferential approach include determining similarities and differences among different domain concepts. And, the hierarchal approach includes pattern understanding and recognition. Ontological representation and visualization of important concept definitions and its relations will be very useful for dyslexic to understand complex types of learning content. Training tests, including clear pictures, diagrams, or visual patterns also will be very useful. In [13], the authors propose a user-friendly mobile learning tool that is based on cloud computing for primary students with dyslexia. This tool provides multimodality functions that are tailored to the different needs of students with dyslexia. Conclusions are that computer-based approach is more helpful for dyslexic students than paper-based approach [14]. Dyslexia adaptive e-learning framework with four

dimensions (Presentation, Hypermediality, Accessibility and user Experiences) is proposed in [15]. This system is intended to provide personalized assessment in an E-learning domain. It also presents an ontology-based approach to develop a personalized e-learning which creates adaptive content based on learner's abilities, learning style, level of knowledge and preferences. Ontology is used to represent the content, learner and domain models. The adaptive learning system is described very briefly, and serious test results are not presented. And proposed ontology models domain concepts as properties, which is not good modeling style.

There are also some good web tools, as specialized E-readers that are an excellent choice to help children with dyslexia. Dyslexia Quest is a tool that can combine fun and education help children with dyslexia. It looks and feels like a game, but it's designed to help children work on various educational elements that dyslexic children typically struggle with.

<http://dyslexiahelp.umich.edu/tools/apps> proposes description and access to more than 200 apps, usable by smartphones that can be useful for dyslexics.

Several systems for supporting dyslexia learners are done and some of them use semantic representation of knowledge. Semantic technologies have sufficient expressive power to represent knowledge in e-learning, but till now they are used only in closely restricted contexts. Well-structured knowledge models that integrate all the knowledge, needed for supporting dyslexia learners in all possible contexts should be developed to ensure the needed level of flexibility, diversity and dynamics of knowledge and learning in this area.

3 Requirements to the model

There are many types of dyslexia and no two individuals will have the same symptoms. So, learning strategies for dyslexics should be flexible, and models, describing learning process and its components should be sufficiently rich. Dyslexia-specific properties of learning resources are related to its presentation, content, language, organization, as well as to its description by metadata. And dyslexics as other learners have also its learning styles, preferences, etc. So, the knowledge model for describing learning for dyslexics should incorporate all the information about learning and the specific one for dyslexics learning needs.

Main requirements to the model are:

- To ensure and organize sufficient knowledge for searching, annotation and recommendation of learning resources in the context of dyslexia learning disability;
- To support combination of dyslexia-related personalization with one, related to other type learner preferences (as Learning Styles, Cognitive Traits, or other preferences);
- To make clear separation between dyslexia-related knowledge and other types of knowledge, needed for resource annotation searching, ranking and reuse;
- To support learner freedom and activity in searching, choosing and using preferable resources for its learning
- To make clear and explicit specific natural language-related learning problems of dyslexics;

- To support collaborative creation, annotation and usage of e-learning resources;
- To ensure interfaces for flexible, attractive and intuitive interaction of all the students with learning software;
- To provide learning guidance and paths based on resource; similarities or learners background (using learner profile)
- To support adaptive personalization according to the learner's changing behavior;
- To provide possibilities to increase user's motivation to learn.

The model also should represent subject domain to meet basic requirements for development of e-learning facilities and determining necessary cognitive abilities for achievement of main educational objectives in the domain. The model of cognitive learning abilities results from mapping of subject domain suitable general pedagogical model is needed to represent dyslexia as a complex of learner's properties. And model that can support reuse of learning resources is needed to guarantee effective and efficient development of learning facilities.

4 Open semantic model of knowledge for adaptive personalized e-Learning environment

The main requirement to successful realization of well-working flexible teaching, learning and resource reuse strategy in the adaptable e-learning environment is to provide machine-processable representation of knowledge about learners, teaching, pedagogical formats, learning domain knowledge, intercomponent communication etc. by usage of semantic technology. This can be achieved if the adaptable e-learning environment is regarded as Knowledge Generation and Management System. We propose architecture that consists of two main components - core (knowledge representation) and shell (application software) component. The core component is responsible for knowledge representation and evolution during development and implementation of e-learning facilities. The shell component covers all knowledge management and learning activities necessary for development of learning facilities, using them and for connection to similar systems through web. It supplies learning and knowledge resources to learners, teachers and resource developers in appropriate manner and organizes learning process.

4.1. Core Component

The core component is represented by knowledge models of two types: Dyslexia-oriented and Dyslexia-independent (see fig. 3). Dyslexia-oriented models contain dyslexia-related knowledge for learners, pedagogy and Document structure model. Dyslexia-independent models describe the main components responsible for development of personalized e-learning facilities: models representing global knowledge about facilities (subject domain, psychological and pedagogical models) and local knowledge (context and communication knowledge models).

4.1.1. Dyslexia-Oriented Knowledge. Main ontologies in this subsystem are Dyslexia Document Presentation Ontology (DDPO), Natural language – related Learning Problems Ontology (NLPO), Dyslexia-related Pedagogical Knowledge Ontology (DPKO), and Dyslexic learner's Profile Ontology (DPO). DDPO represents all concepts that are used to characterize learning objects (LO) (learning facilities) as relevant to individual dyslexic learner. It has clear structure with explicitly stated learning goals, parameters concerning LO presentation and all knowledge needed for applying layout annotation approach to extract structural information from the document and creation of interrelations between distinct parts, which eventually complement the previously extracted information.

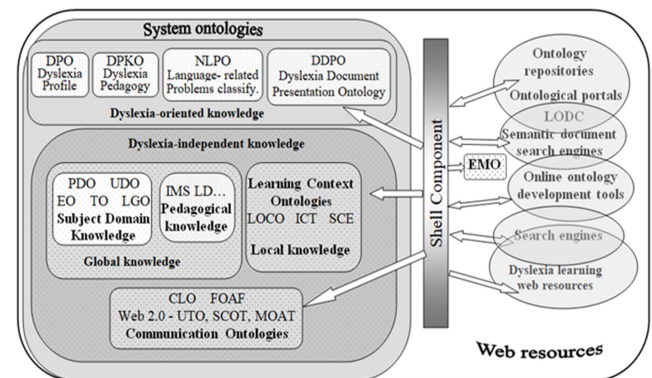


Figure 3: the Open Semantic Model of knowledge for Adaptable personalized E-Learning Environment

The NLPO classify learning problems of dyslexics, related to natural languages. This ontology can be used for easy adaptation of the system to different natural languages. Some languages, like French, Danish and English, are more hard for students with dyslexia, and others like German and Italian are easier. For example, in some languages it is harder for learners to hear some sounds than in others. Spelling in Spain is predictable, but, this is not the case for English. NLPO classify difficult language elements for every natural language.

The DPKO classify specific educational objectives and learning goals fixed in a teaching model for dyslexics. The dyslexia-related teaching model is a goal-oriented model derived from the general model of subject domain and general teaching model that includes objectives and goals, related to dyslexia problems. Learning goals determine concrete learning activities for achievement of concrete result as comprehension of symbol-phoneme correlation for example. Dyslexic Learner's Profile contains information about learner features and its learning characteristics needed for personalization.

The DPO includes all valuable (dyslexia related or some unrelated) learner's profile aspects: personal, personality, some learning styles and preference data [16]. Personal data comprises short biographical information about a learner (mainly for mapping to other profile ontologies). The personality data is for capturing individual behavior in the learning process: psychological type, concentration capability, attention capabilities, etc. Learning styles are for ensuring mapping of dyslexia learner

profile to other general psychological, pedagogical or competence ontologies, used in e-learning.

The preference data stores a set of student preferences regarding the system customization (preferred presentation format, language for content display, sound and video characteristics, etc.) The learner model is partially open for changes and mappings for dyslexia-independent learner ontologies are important.

4.1.2. Dyslexia Independent Knowledge. Dyslexia independent models represent global and local knowledge about learners, learning content and communication aspects concerning the development, searching, ranking or usage of personalized learning facilities. The global knowledge is represented by subject domain model, general learner model and pedagogical knowledge. Local knowledge model represents local context of the learning process. The Subject Domain Model is formal representation of subject domain knowledge intended for learning. It is basis for construction learning goal-oriented model of subject field. The learning goals concern the main concepts (knowledge) that have to be acquired. This model consists of Precise Domain Ontologies (PDO) and Upper Domain Ontology (UDO) and mappings between these ontologies. The UDO describes general domain concepts and related to them external domain concepts. The PDO presents all details of the domain knowledge. Specific PDO is used for every subdomain. All these ontologies can be stored or retrieved from libraries. There is also small specific Etalon Ontologies (EO) for automatic generation of testing resources. The learning goals achieved by each LO is represented by little ontology that describes the essence and relations of the underlined subject knowledge. It serves for learning goals organization in Tracing Ontology (TO) using for determination of learning goals achievement path.

Pedagogical knowledge model represents subject-independent pedagogical and psychological domains (learning goals, learning standards, etc.). Its basic conceptual elements are pedagogical methods, knowledge delivery modes and psychological factors for efficient learning. The methods correspond to the basic learning approaches - content-centered, activity-based and collaborative learning. Each delivery mode presents a way of effective implementation of pedagogical methods. For example lectures and tutorials are knowledge delivery modes realizing the content-centered method. The factors for efficient learning influence achievement of desired learning results. Some of these factors concern attention, motivation, self-assurance, emotions and knowledge retain of learners. This can be used for development of pedagogical domain ontology (PDO).

General Learner model includes dimensions of individual differences such as cognitive styles, learning styles, level of knowledge, age, gender, that are not related to dyslexia. The learning preferences are related to learning competence and learning environment. The learning competence is determined by cognitive abilities, learning skills and educational background of learner. The learning style information includes data presenting these features. This data describes learner's behavior in learning situations. Several learning style models for example has been proposed in the literature [17], including Myers Briggs theory of Learning Styles, Kolb Learning Style Learning Style Inventory, The Ned Herrmann Whole Brain Dominance Theory, and The Gregorc Style Delineator [18]. Every of them should be described semantically (by ontology) and mappings between these

ontologies should be presented. Every concrete implementation of our model can use one or more of these learning style ontologies.

Local knowledge model represents context ontology that includes Socio-Cultural Environment (SCE) ontology, Information and Communication Technology (ICT) ontology and Learning Object Context Ontology (LOCO). The SCE describes concepts related to social and cultural learner's features. The ICT ontology serves for conceptualization of technological basis of learning facilities development. LOCO is developed to promote integration and reuse of LOs and learning designs. In our e-learning environment learning objects correspond to learning facilities that is considered as learning resources (LR), described by some learning standard.

Communication model includes Web 2.0 communication ontology. Collaborative Learning Ontology (CLO) facilitates shared understandings of several early presented models of collaborative learning and collects rational models of effective collaborative learning [19]. It is taxonomy of concepts, representing collaborative learning sessions and interaction among learners during it. CLO offers a general framework and vocabulary to describe collaborative learning scenarios based on theories and standard vocabulary and knowledge that can be used by pedagogical agents in their communication and negotiation. Collaborative learning and authoring is based on social tagging, bookmarking, blogging and instant messaging. The SCOT ontology [20] represents structure and semantics for social tagging data and provides methods for sharing and reusing them. The MOAT ontology [21] provides a model for definition of relationships between tags and their meanings. The FOAF [22] is an ontology describing persons, their activities and relations to other people and objects. It describes social networks on base of grouping.

Well-structured knowledge in subdomains in e-learning should be represented by several mapped ontologies. So, multiple, interoperable ontologies should be used to characterize learning process from different perspectives. Different ontologies can be used in different concrete systems, based on this model according to the specific characteristics of participants in the learning process and main learning goal. Modular organization of knowledge ensures flexible usage. A grand variety of specific ontologies can be stored in libraries and only needed ones should be selected, mapped and used in every specific system for learning.

4.2. Shell Component

The shell component provides two groups of functions: knowledge management and knowledge exchange. The former includes archiving, collection and modification, the latter – transfer, access, store and usage. This component uses External Model Ontology (EMO) that bases on main elements of the ontology concerning the core knowledge models and additional web resource characteristics as server, service characteristics, license agreements, domain dependency, resource URLs and valuable metadata, systematized in external ontologies. This information is useful for practical realization of the main modern e-learning principles: personalized collaborative learning, cooperative authoring, learning resource integration and reusing, dynamic management of personalized learning resources. Learners' personal agent uses this information for searching useful web based (collaborative) learning assistant and other useful web learning recourses. On the base of EMO domain experts or software tools find useful web ontologies or ontology

management tools. The teachers can find also web graphical ontology editors, learning repositories, web-based authoring tools and the latest pedagogical or subject domain knowledge. The shell component of the KGMS puts the (re)use strategy into effect. We suggest an agent-based approach for realization of this component.

5. Agent-Based Framework as a shell model

To ensure sufficient level of flexibility, needed in dyslexia learning and support resource reuse, we present an open agent-based framework that supports learning facilities production, composition and use, as well as management, (re)use and retrieval of external recourses that are accessible through cooperation of intelligent agents (fig. 4). Knowledge expert agent provides tools for resource developers. It interprets expert query and delivers needed learning or knowledge-specific resources (in some cases using communication with other internal or external agents). For example, when knowledge engineer develops learning resource, he could use his expert agent to invoke external resource or search in the web agent for some related to the topic materials using EMO. When he works on ontologies, expert agent can invoke ontology search agent or mapping agent to perform needed services.

Teacher agent directly communicates with educators. It interprets educators' query, invoke authoring or learning process control tools in necessity or delivers the needed resources communicating with other internal or external agents. For example, if a teacher need of a resource that is not in the local e-learning system the agent sends a query to an external resource agent and waits for the results to present them to the teacher. Then the teacher can request the annotation agent through his interface agent to annotate returned results according to UDO, PDO, DDPO, IMS LD ontology [23] or other ontologies, then invoke evaluator agent and choose the best ones according to the returned evaluations.

Dyslexic learner personal agent communicates with a learner. It services dyslectic pupil searches for local learning recourses or sends a request to an external resource agent, if the learner has specific needs, which the e-learning environment can not meet. This agent observes the learner's behavior and collects information related to the dynamic user profile components in DPO. In such a way it periodically updates its personal information after any change in learner's interests or competences. It also, creates, or follows learning paths, recommends appropriate learning resources to the learner after comparing the information about the learner and resources.

Collaboration agent collects and manages information about communication (in communication ontology) of users and tagging. On this basis it makes group characteristics-based recommendations about resources. It is permanent active and acts as server that provides instant message services and can recommend to dyslexic learners other learners with similar learning disabilities for help on base of information about their affective aspects, social behavior and learning outcomes. Evaluator agent is invoked by the teacher agent to observe and evaluate the learner's behavior and results. It automatically extracts personal information about learner's competences during learning or assessments and inserts it in profile ontology.

Resource agent (Dyslexia) is intended to analyze and recommend suitable resources to the learner having specific dyslexia problems.

Ontology search agent collects information about ontology web repositories and semantic document search engines, and stores it in EMO.

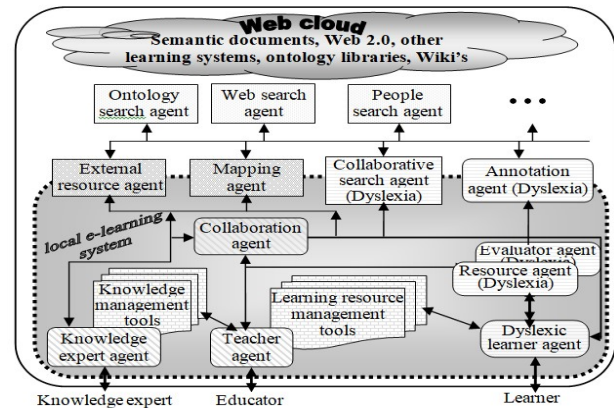


Figure 4: The model of our agent-based framework

It possesses information about ontology search query (and eventually some additional information about searched ontology). It uses this information in dependence on stored in EMO knowledge about the web ontology for searching in ontology repositories or sending query to semantic web search engine and for filtering the received results.

Web search agent collects information about learning resources repositories and web search engines, stores it in EMO and in dependence on the search query usage sends queries to web search engine and filters the received results.

Mapping agent maps the received web ontologies to local domain ones (to estimate web ontology relevance), or folksonomies (from web 2.0 external resources) to domain ontology. The mapping is used for improving accuracy of web search results, as well as for adding new ontologies in the system, given from the library or from interned. To guarantee interoperability between learning resource repositories in search, use, and composition of new resources, it is needed metadata specification by ontology in each repository. The agent uses its mapping strategies to map repository ontology dynamically, when learners or teachers search learning resources in web repositories.

Annotation agent generates automatically learning resource metadata related to learning disability or learning domain. Both resources, developed by teacher or ones, found in the internet should be annotated. The annotation agent has three main tasks: ontology-based metadata generation, collaborative tagging-based annotation and annotation based on statistical algorithms.

External resource agent (coordination agent) receives messages from the assistant agents and takes decisions to which agent(s) to forward them. It organizes the communication between the agents and guarantees easily extension of the system with new functionality agents, if it is necessary.

Learning profile and some other ontologies also can be stored on the cloud, combined with social media at a macro level and learning analytics/big data can be used as a basis for automatically providing recommendations. For adult learners a portfolio also can be used. The MAS can also identify skills and knowledge gained during any social media interactions.

The advantage of Intelligent MAS is that it can dynamically act on the correspondence with learners cognitive and motivation

size) can be stored during development both in the recourse(as metadata), and in the recourse ontology. This information is automatically extracted from resources by Annotation agent or manually added by developers and is important for resource ranking and recommendation in correspondence with specific criteria.

The screenshot displays the Protege 3.4.4 interface with three main panes. The left pane shows the 'Asserter Hierarchy' for 'owl:Thing', listing various academic skills and learning disabilities. The middle pane shows the 'PROPERTY BROWSER' for 'Has_Learning_Style', displaying a list of properties such as 'Has_Adapted_audio', 'Has_domain', 'Has_formatting_style', 'Has_language', 'Has_layout', 'Has_style', and 'Use_learning_standard'. The right pane shows the 'PROPERTY EDITOR' for 'Has_Learning_Style', displaying the 'rdfs:comment' property.

owl:Thing Hierarchy:

- owl:Thing
 - assessment_of
 - dyslexia_type
 - formatting_style
 - Letter_recognition
 - Reading_Disorder
 - Reading_fluency
 - Reading_speed
 - Word_recognition
 - Writing
 - Functional_Skills
 - Intellectual_Disability
 - Learning_Disability
 - ADHD
 - Aphasia
 - Discolouia
 - Dysgraphia
 - Dyslexia
 - Deep
 - Developmental
 - Information_Process

PROPERTY BROWSER: Has_Learning_Style

- Object Datatype A
- Object properties
 - Has_Functional
 - Has_Intelector
 - Has_Learning_Style
 - Has_Dyscolou
 - Has_Dyslexia
 - Has_Motivation
 - Has_Skill
 - Has_Math_S
 - Has_Readin
 - Has_Social_Writing
 - Has_Social_P

PROPERTY EDITOR: Has_Learning_Style

- Property
 - rdfs:comment

Figure 5: Some ontologies in our knowledge model

We evaluate our resource agent’s capability to examine resources presentation quality by using it to compare resources, downloaded from web sites for dyslexia learners and analogical resources from other web sites. The resource agent extracts from given resources parameters, (some of them are shown in table 1 and table 2) and calculates the average value for applicability of the resource for dyslexia learning, using values, shown in tables.

Presentation parameters can be grouped in two main groups: having soft –bad values and hard - bad values.

Soft-bad values parameters can't make content unusable (examples are given in table 1)

Table 1: some soft (good and bad) values

Align type	value	underlining	value	italics	value	upper case	value	Hyperlinks look	value	Glossary	value	section headings	value	table of contents	value	Voice	value
left	1	yes	0	yes	0	yes	0	different	1	Yes	2	Yes	5	Yes	5	active	1
others	0	no	1	no	1	no	1	The same	0	No	0	No	0	No	0		0

Using columns:	coefficient	Abbreviations	coefficient	text color	coefficient	Background color	coefficient
one	1	All	0	dark	1	dark	0
two	0.5	Standard	0.5	light	0.5	light	1
More	0	avoid	1	Red/pink	-1	Red/pink	-1

Hard-bad values parameters can make content unusable (examples are given in table 2)

The given very large or very small values of hard-bad values parameters are closely-related to the importance of these parameters and lead to very bad values of calculated average measures for usability of corresponding resources by dyslexics.

Table 2: some hard (good and bad) values

Body text	Font	value	Headings Font size	value	character spacing	value	word spacing	value
12, 14		10	14-16	5	Larger 30%-40%	5	3.5 times letter spacing	10
11, 15		5	13, 17	2	Larger 20% or 50%	2	4 times or 2 times	5
10, 16		0	12, 18	1	Larger 10% or 60%	1	5 times or 1.5 times	2
9, 17		-2	11, 19	0	Larger 60% >	0	normal	0
8, 18		-4	10, 20	-2	normal	-1	6 times or 0.8 times	-2
<8		>0	<8	-10	smaller	-10	< 0.8 times	-10

We try some combinations of values of parameters, shown in the tables, and some other important parameters during the experimental evaluation. The combinations, ensuring the best classification results are shown in tables. Machine learning algorithms can be used to find the best values of such parameters. The size, presentation and content of the learning objects should correspond to the learner's cognitive properties that can vary significantly for different dyslexia learners. So, learning environment for dyslexics should be extra flexible and adaptable. We believe that presented knowledge and agent-based architecture is the most appropriate as a basis for development of e-learning systems for dyslexics. It is also important to use simple and adaptable interfaces for interaction with learners, and our architecture can support easily development of such ones. The system also proposes learning objects (ontologies) to improve the understanding of learning content. The domain ontologies can be used for graphical representation of content structure and this will support understanding. High quality and easily accessible learning resources will also support and increase motivation of dyslexics to learn.

7. Conclusion

We present flexible, scalable model for personalization of e-learning for dyslexics that aims to improve quality of learning by adapting the system to meet individual needs and preferences of every learner. The system uses mapped ontologies to collect and systematize valuable information, needed for personalization. Clear separation of dyslexia-related and dyslexia-independent knowledge make our model very flexible, customizable and adaptable to learning for peoples, having other type disability or learning difficulties. This model can be used to develop learning systems for learners, having or not some disability. The main difference between different such systems will be in the selected or mapped ontologies in its knowledge base. And agent-based architecture makes the system highly adaptable to the needs of every learner. Every agent can be modified without affecting its environment (other agents or ontologies) including collaborating between agents. Other new agents also can be easily integrated in the system. Ontologies also can be changed or new ontologies can be added if necessary.

This platform for development and usage of personalized e-learning facilities for dyslexic learners is a step to ensure effective and efficient adaptable e-learning environment. The main drawback of ontology-based modelling of generated knowledge and e-learning resources is difficulty and complexity of knowledge acquisition and its representation as ontology. We believe that clear and possibly standardized description of small sub domains (as learning styles, learning standards, dyslexia, etc.), using small ontologies, and providing mappings between these ontologies will simplify modeling of complex domain by supporting ontology reuse.

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